

Crown and bridge

Post crown preparation

LECTURE 9

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Post crown
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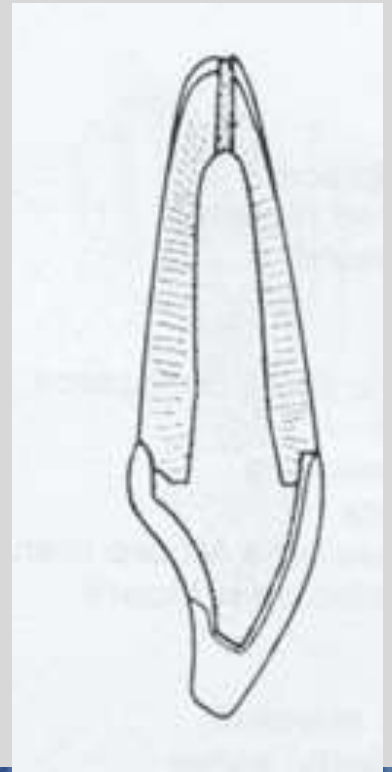
Post crown

Post crown

It is a fixed restoration which replaces the complete coronal portion of a tooth and is retained by means of post (dowel) which extended into the root canal.

The post crown (1) reinforces the remaining tooth structure against applied forces by distributing them inside the root.

Some studies said that post not only distribute the forces applied from occlusion but (2) provide retention for crown.



Factors influence the choice for post crown:-

1-The type of tooth (whether it is an incisor, canine, premolar or molar).

2-The amount of remaining tooth structure. Which is the most important factor which determine the prognosis of the crown.

Before preparation, existing endodontically treated tooth need to be assessed for the following:-

Good apical seal (radiograph).

No sensitivity or pain to percussion (clinical exam.).

No exudates, No fistula.

No active inflammatory signs and symptoms.



Indications:

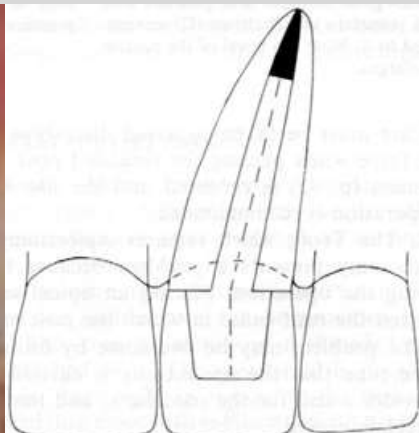
Restoration of endodontically treated teeth that have:

- 1- excessive amount of the tooth structure is removed or lost by caries or trauma making the retention of other types of crowns impossible(the remaining tooth structure is unsuitable).
- 2- Core construction is needed.
- 3- Intra-canal retention is the only mean of retention possible.



Indications:

- 2-Realignment of malposed teeth.
- 3-As a single short span bridge retainer(cantilever bridge).
- 4-Tooth with short clinical crown.



Factors to be considered in selection of a tooth for post crown

1-The root of the tooth should be sufficiently shaped, with adequate length and width:

- * If there is an abnormal shape of canal (curved roots or canals) post crown will be contraindicated.

- * Length of the root should be sufficient to be used for post crown.



2-The root should be without internal or external resorption.

3-The alignment of the root in relation to the adjacent teeth must be suitable for insertion of post crown.

4- The root should be filled with well-condensed gutta percha or any other soft filling materials especially at the apical third of root space.

5- occlusal relationship Should be evaluated.

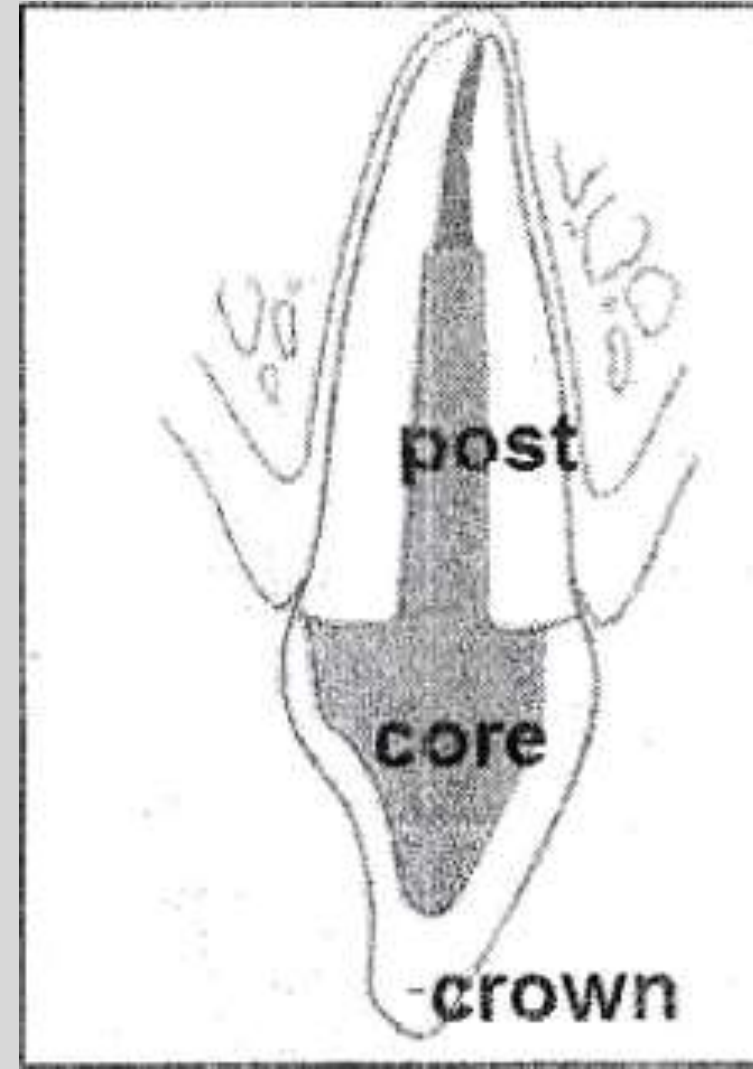


Basic components of post crown

A) Post (dowel): the part that extended into the root canal and gives support and retention for coronal restoration.

B) Core: It is the coronal extension or addition to the post necessary to provide the desired retention for the restoration.

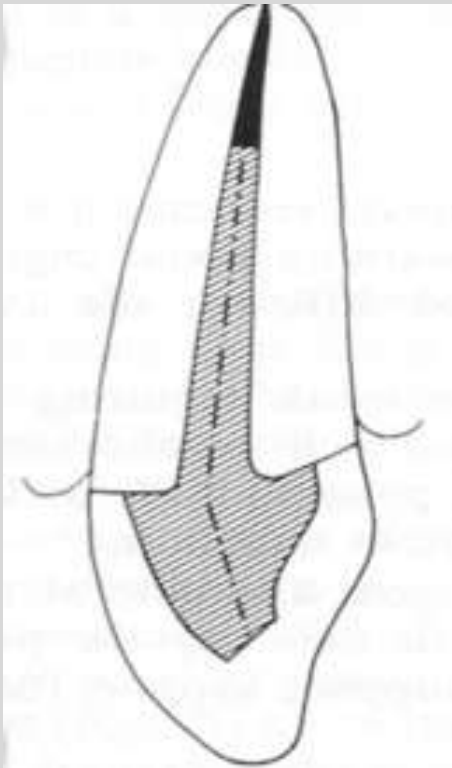
C) Crown: It is the restoration that placed on the core, it may be full-metal, jacket crowns or as retainers for bridge



Types of post- crowns

1-Two units post crown (post and core +crown):

It is commonly used type especially with ready-made posts (screws), and when tooth alignment is not suitable with other bridge abutments.



2- one unit post crown:

it is used as a single crown or as retainer of a bridge when the root is in good alignment with other abutments.



One-unit post crown

Advantages and indications of two units system post-crown

- 1-** Young patients under 18 years old, because the gingiva-tooth relationship will change with time.
- 2-** The two unit system can be repaired if crown is damaged.
- 3-** When the endodontically treated tooth is to be used as a bridge abutment, it is not necessary to do the post crown preparation parallel with the other adjacent abutment teeth.
- 4-** Marginal adaptation and fitness of crown restorations are independent of the type of dowel used.

POST-CROWN PREPARATION:

Procedures:-

A three – stage operation:-

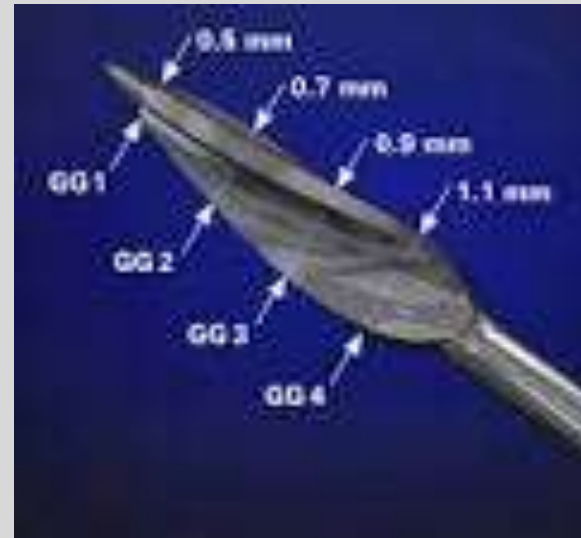
- 1-Removal of the root canal filling material to the appropriate depth.**
- 2- Enlargement of the canal.**
- 3- Preparation of the coronal tooth structure.**

A post cannot be placed if the canal is filled with silver point which must be removed and the tooth retreated with gutta-percha.

Before removing G.P. calculate the required length of the post

As a guide, make the post length equal to:-

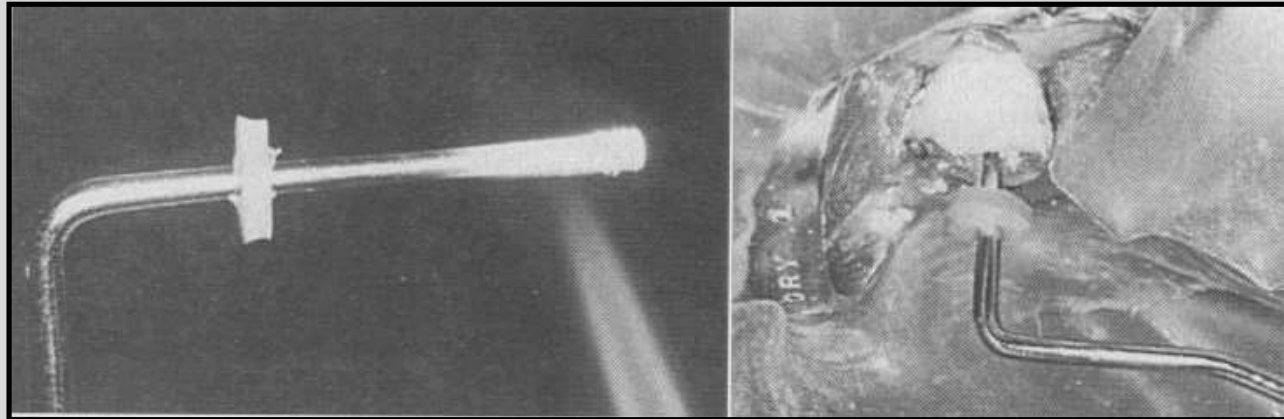
- 1- The height of the anatomic crown**
- 2- Two – thirds the length of the root.**
- 3- leave 5mm of apical gutta -percha. On short teeth, An absolute minimum of 3mm of apical fill is needed.**



1- Methods for removing Gutta Percha:-

A – Using warmed endodontic instrument (plugger)

Select large plugger that holds heat but not binds against the canal walls.



B- Using a rotary instrument:

By using (safe-tip) instruments because they are not end-cutting burs.

The friction generated between the filling and the tip of these burs softens the gutta percha. Peeso-Reamers and Gates Glidden drills are often used for this purpose.

End-cutting instruments should never be used. The rotary instrument should be slightly narrower than the canal.

Make sure the instrument follows the center of the G.P. and does not cut dentin.

Gate-glidden



2- Canal preparation:

When the G.P. has been removed to the appropriate depth, shape the canal as needed, by using an endodontic hand instrument or a low –speed drill.

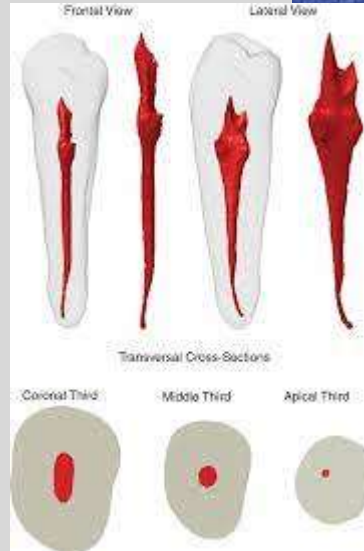
Knowledge of average root dimensions is important, the minimum root wall thickness is 1mm.



2- Canal preparation:

Knowledge of root canal cross section also is significant in post selection.

Prefabricated posts are circular in cross section, but many root canals are elliptical, which makes uniform reduction with a drill is difficult.



Post & Core



Post can be classified into two main types:-

1- Custom-made post: casted in metal, indicated for teeth with root canals whose cross section is not circular or is extremely tapered.

2- Prefabricated(ready-made) post

A-tapered, smooth-sided posts,

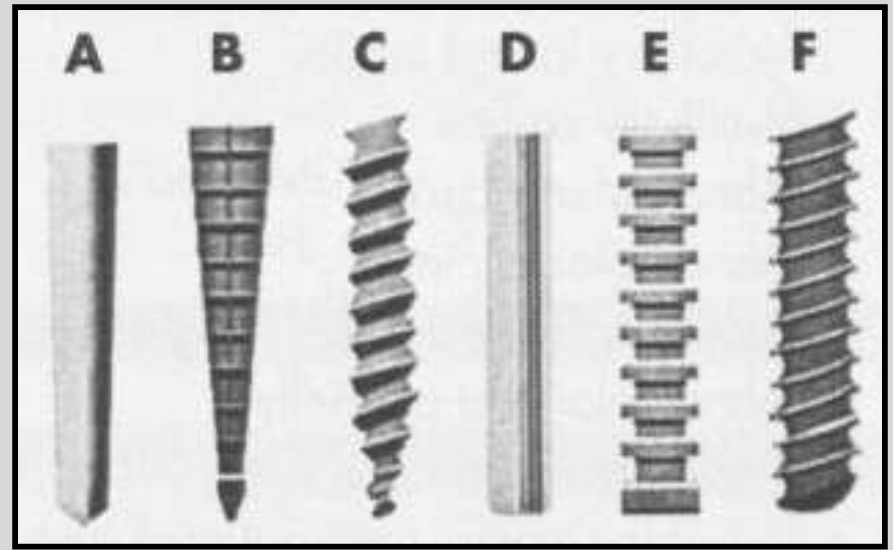
B-tapered, serrated posts

C-tapered threaded posts

D-parallel, smooth-sided posts

E-parallel, serrated posts

F-parallel, threaded posts



Prefab. Posts fabricated from different materials:-

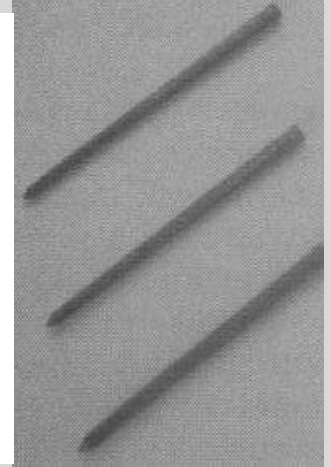
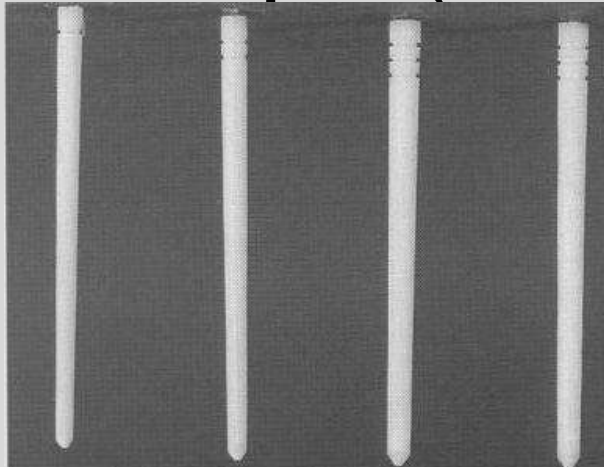
Posts made from precious, semiprecious, and non precious alloys.

Carbon –fiber posts gave increased popularity in recent years consist of carbon fibers embedded in an epoxy matrix; it is strong with lower stiffness.

One advantage of carbon fiber post is the ease of its removal by drilling epically.

The chief disadvantage of a carbon fiber post is its black appearance.

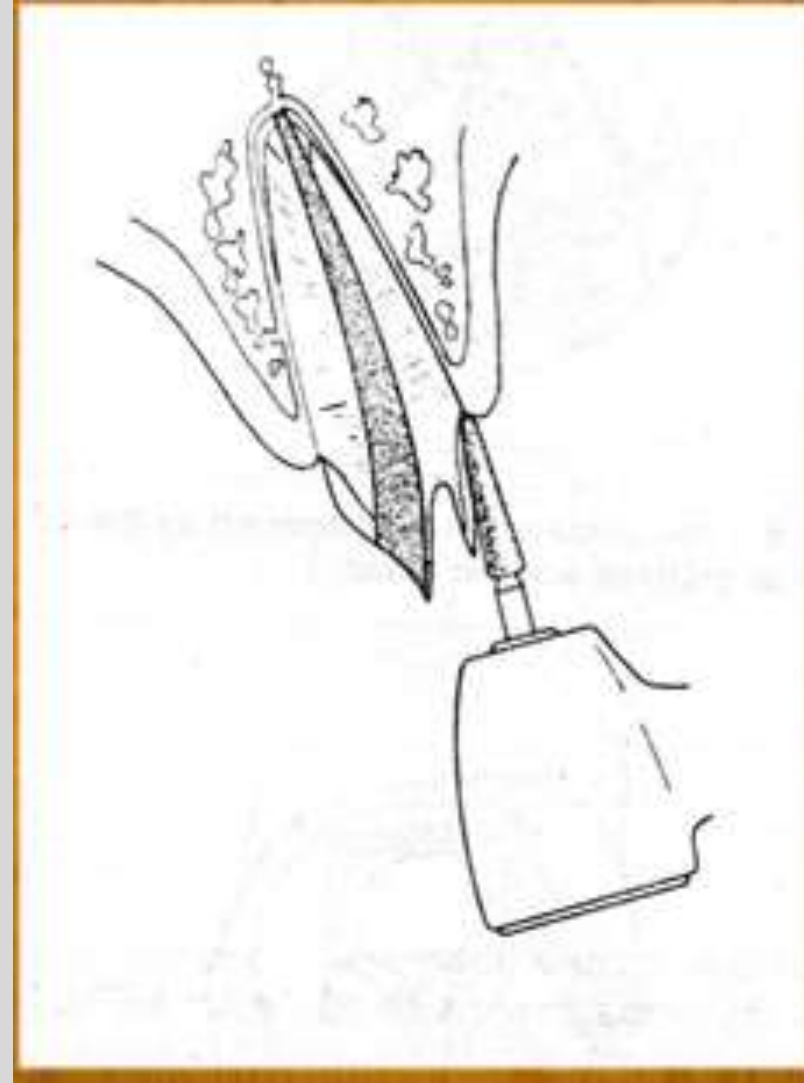
High strength ceramic posts (Zirconium) have excellent esthetic properties.



Preparation steps:-

Preparation of coronal portion:

In a weakened endodontically treated tooth we should remove any undercut, unsupported enamel, previous filling, cement base, and any weak part which may fracture later on leaving 2-3 mm. sound tooth structure supragingivally.



Preparation of coronal

Notes: portion:

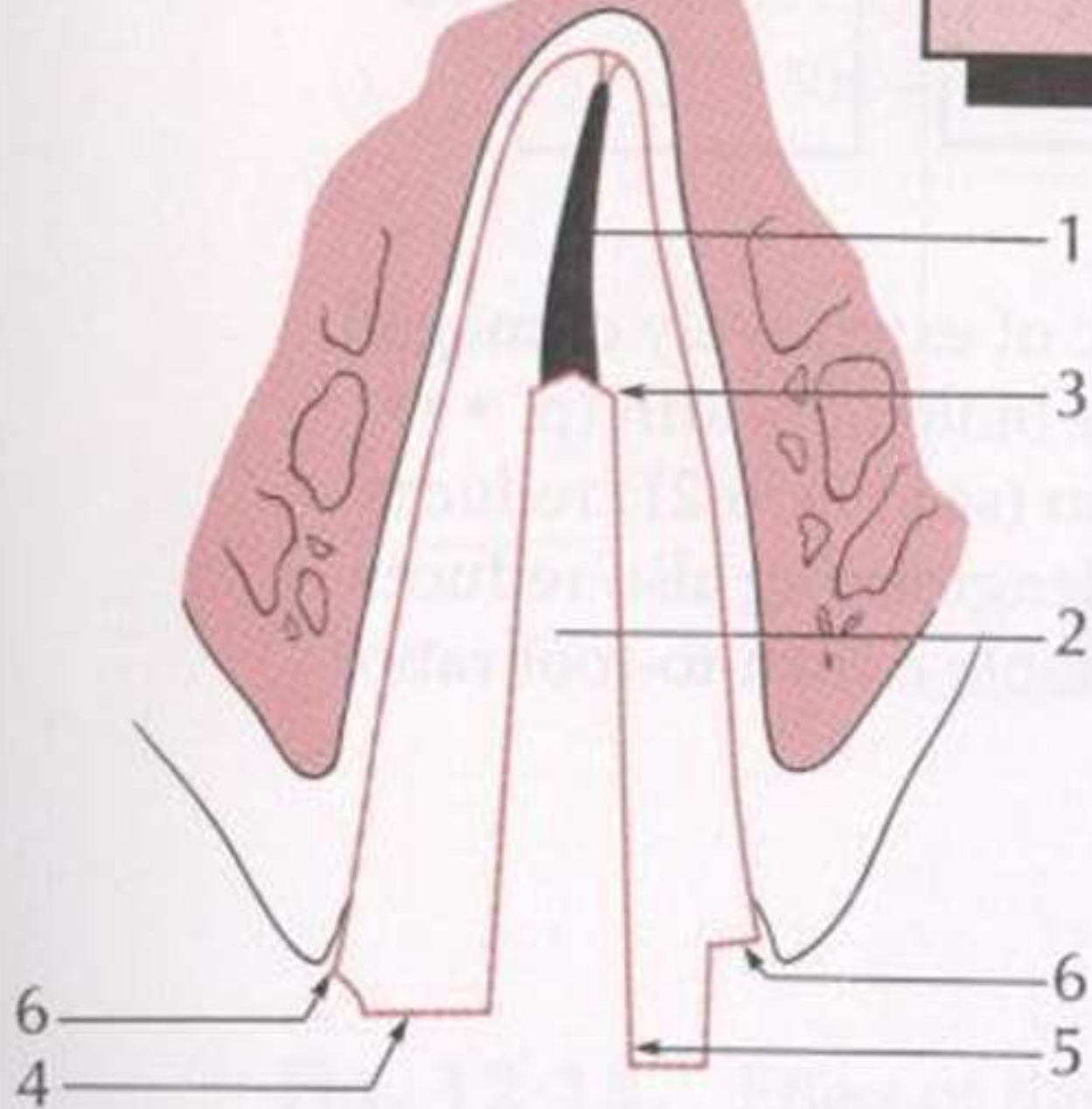
Be sure that the facial part is adequately, reduced for good esthetics (enough reduction with shoulder finishing line).

Wall height is reduced proportionally to the remaining wall thickness because tall, thin walls have a tendency to fracture during try-in and seating of the casting.

Be sure that part of the remaining coronal tissue is prepared perpendicular to the post, because this will create a positive stop to prevent over seating and splitting of the tooth.

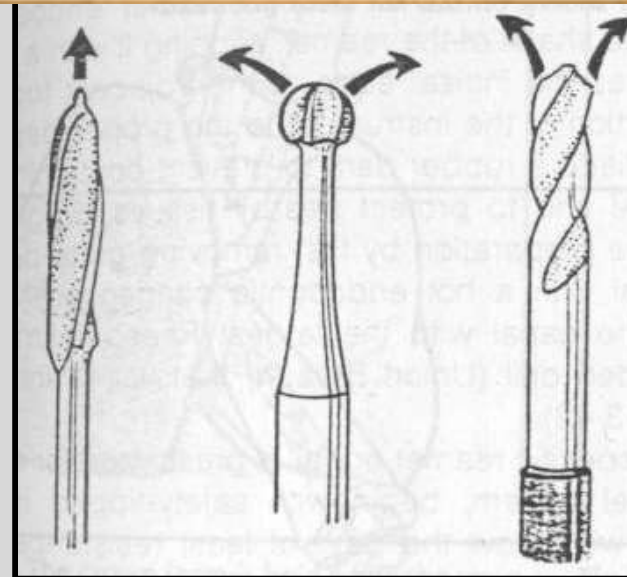
Complete the preparation by eliminating sharp angles and establishing smooth finish line.

• Mar



Root canal preparation:

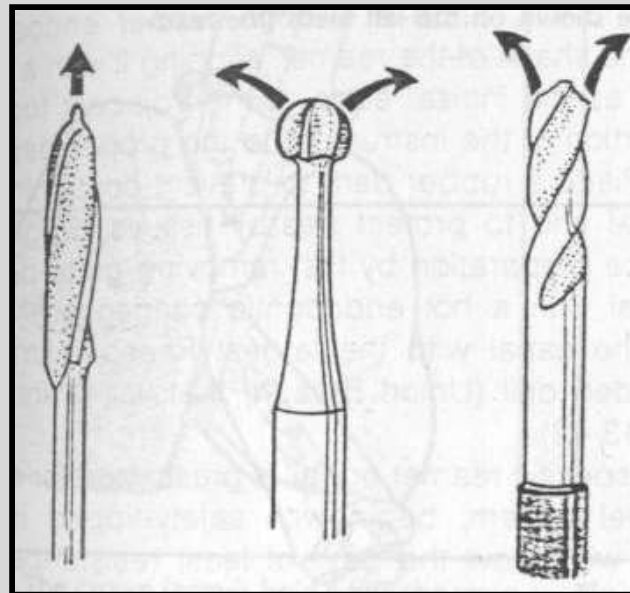
The bur used to prepare the root canal is called "peeso reamer" or "peeso bur" it comes in different sizes and the size used depends on the size of the root canal, it has a blunt non cutting end which will follow the area of less resistance (gutta percha) without perforating the root.



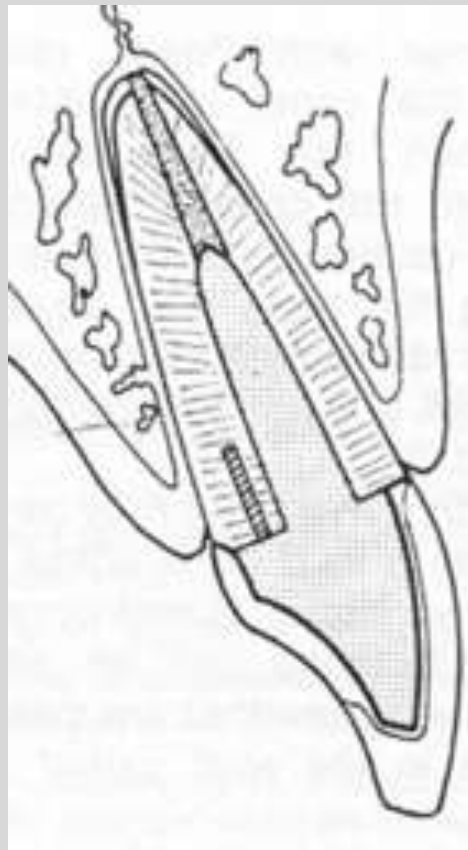
A) A radiograph is taken to evaluate the length, width, shape of the tooth and the type and quality of the filling especially in the apical third of the root canal.



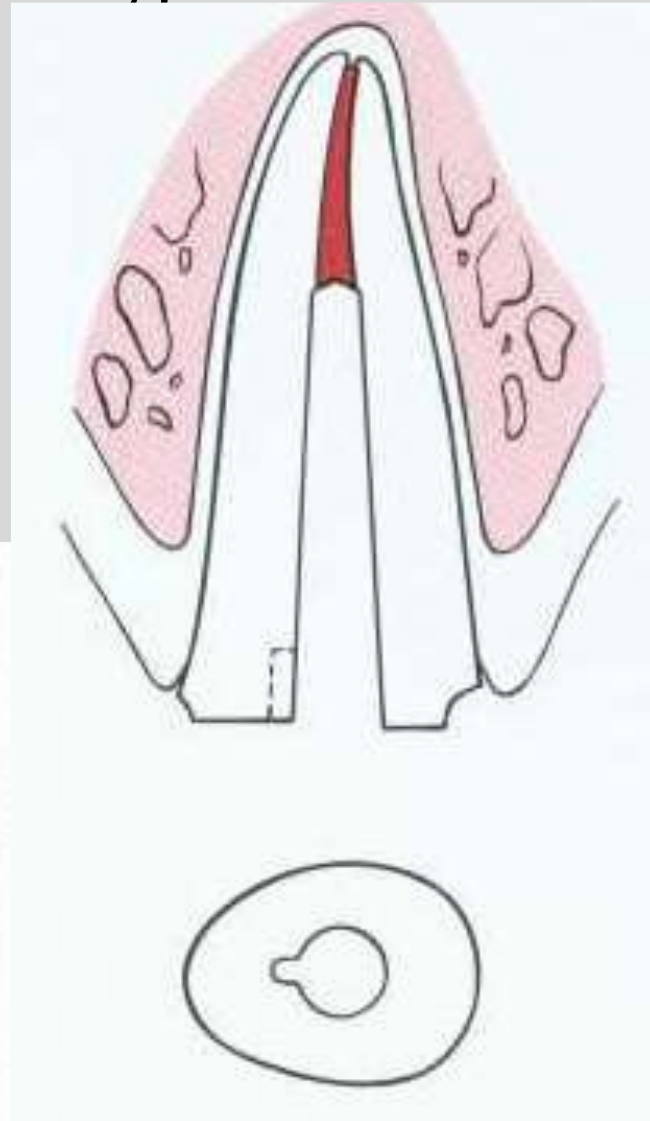
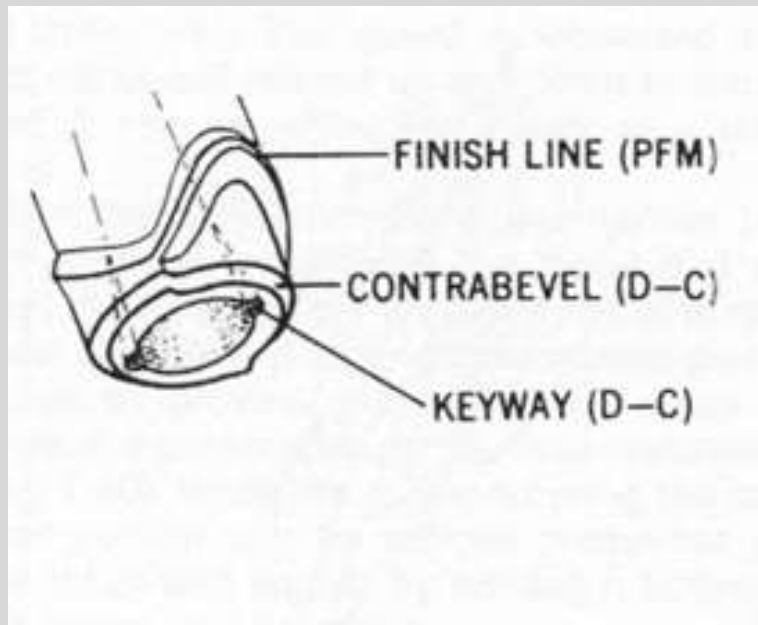
B) With the use of peeso bur, the root canal filling material (gutta percha) is removed up to $\frac{2}{3}$ of the root length leaving 3-5 mm. filling at the apex, the space made by the removal of the gutta percha is for the insertion of the post which should be as long as possible to get maximum retention and at the same time prevent movement of the apical filling material which if it happens it will lead to leakage and failure.



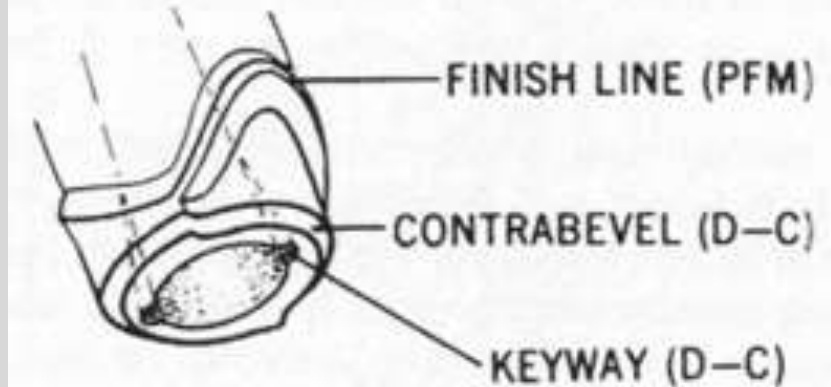
C) The canal should be parallel sided with slight flaring to the outside, in short teeth accessory retention means may be used as pins, where the pin hole should be placed parallel to the post canal preparation.



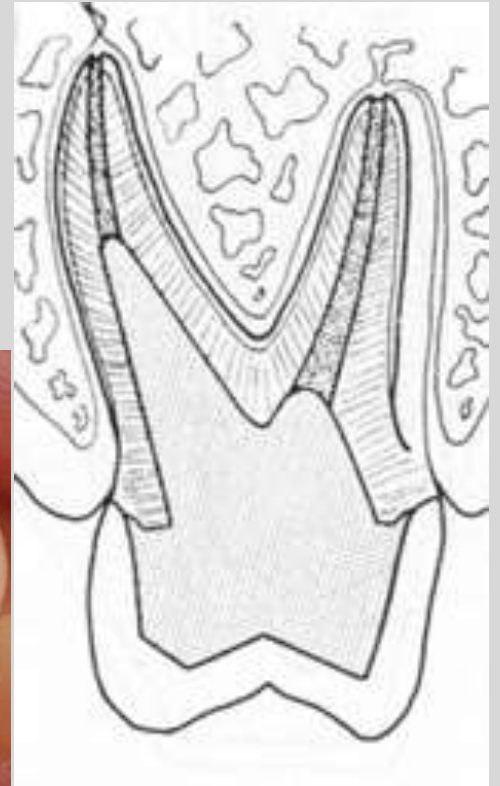
D) A keyway is made at about 1 mm. depth, and 4 mm. length extended to the canal using a flat end fissure bur, this will act as a guide during placement of the final post, and will prevent the rotation of the post especially with teeth with a round cross sectional root canal.



E) Contra bevel, it is a reverse bevel placed around the occlusal external surface of periphery of the preparation, this will facilitate placing a collar(a metal thickening) around the occlusal periphery of the preparation which will help in holding the tooth together and preventing the fracture of the remaining tooth structure.

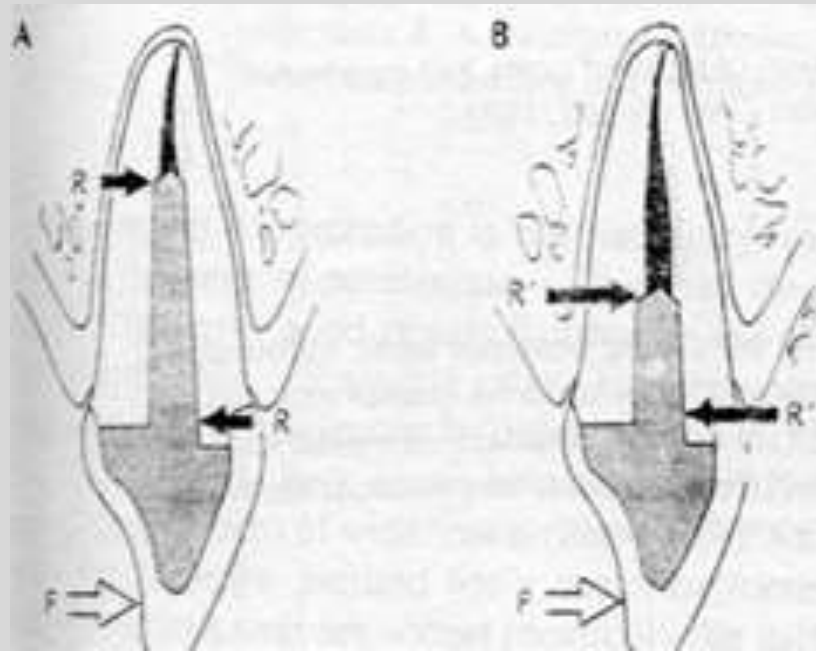


F) For multicrooted posterior teeth, the post should be placed in the largest canal, EX: palatal root for the maxillary molars, and distal for mandibular molars. Placing the posts in all canals might weaken the tooth, and they might not be parallel to each other. A short small second post may be helpful in preventing rotation and increase retention.



Retention of the post crown depend on:

- 1-Taper of the root canal, as the taper increase retention decreases.
- 2-Post length, as the length increases retention increases.
- 3-Post diameter, increasing the post diameter is not recommended because it weakens the tooth.
- 4-Post surface texture, a post with rough surface is more retentive than post with smooth surface.



How to prevent post rotation

1-Keyway.

2-Triangular shape canal in incisors and elliptical shape for upper canine.

3-Pins.

